Zhenbang (James) Xie
University of Pennsylvania | E-mail: xie10@seas.upenn.edu | Telephone: (+1)8563284348

Education

North China Electric Power University (NCEPU)
Bachelor of Engineering, Automation
Beijing, China | Graduation: June 2025

Purdue University Northwest (PNW)
None degree program, Electrical and Computer Engineering
Hammond, IN, USA | Duration: Spring, 2024

University of Pennsylvania (UPenn)
Master of Engineering, Electrical and System Engineering
Philadelphia, PA, USA | Expected Graduation: June 2027

Technical & Analytical Skill

Control systems design
3D modeling
Engineering drawing
Thermal theory and fluid mechanics
Circuit design and analysis
Digital system design
Boiler and instrumentation in thermal power plants
Renewable power systems (wind and PV)
Process control

Programming & Software

C/C++, Ladder Diagram Language (PLC programming), ETAP, MATLAB, AutoCAD, NI Multisim, Arduino

Professional Experience

China Energy Chengdu Jintang Power Generation Co., Ltd. — **Chengdu, China**
Thermal Maintenance Intern — *July 2024-August 2024*

Collaborated with the thermal maintenance team to gain experience in coal-fired power generation. Observed and learned from senior engineers about system maintenance, sequential control processes, and control cabinet design, gaining hands-on skills and technical knowledge in power plant operations.

Leadership & Project Experience

- **Renewable Energy Power Plant Design Project** — **PNW**
  Led a team in designing a renewable energy power plant, including traditional power plant, wind farm, and PV farm, using ETAP, demonstrating project management and technical leadership. Got full score from the professor.
- **PID and Fuzzy Control System Design** — **NCEPU**
  Designed and implemented PID and fuzzy control systems to manage superheated steam temperature in coal-fired power plants. Took initiative in project planning, troubleshooting, and testing.
- **Frequency Response and SFD Inhibition Methods of Wind Power Generation** — **NCEPU (Senior design)**
  Developed a MATLAB/Simulink simulation model for the grid-connected frequency response of a PMSG-based wind power system. The model has wind turbine and PMSG dynamics, a back-to-back dual-PWM full-scale converter, a DC-link stage, machine-side and grid-side converter control, and virtual inertia control. Based on the model, I simulated frequency responses under load disturbances, analyzed the secondary frequency drop during rotor speed recovery, and evaluated an adaptive droop control strategy under different wind power penetration levels.
- **Waldo Project** — **UPenn**
  Built a robot with two mini servos in different degrees of freedom, and a box with two potentiometers to control the servos respectively. An ItsyBitsy ATmega32u4 development board was used as the signal processing part of the system. The project was part of the course project of Mechatronics. Demonstration link:
  https://youtube.com/shorts/pjFLa3Hlalg?si=dqxOPCthK5ngL253
- **Robotic Tank** — **UPenn**
  As a group of three, we build the vehicle from mechanical design to software implementation using ESP32-S2. As the leader of our group, I am taking responsibilities from assisting mechanical design, testing electronics and designing the controller of our vehicle. Our robot is capable of wall-following using three ToF sensors, has good manual control performance, and wins the course final robot competition. Demonstration link:
  https://drive.google.com/file/d/1kZ_fjWkYxZZY3IzEVh8XaS_SS7Zcgd_E/view?usp=drive_link
  https://drive.google.com/file/d/1-jsmzIuyaKlhYmkqx9QGhBg4kG26P-_B/view?usp=drive_link